Class Test 4 January 2025

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-4/T-1 CSE 409: Computer Graphics

Time: 25 minutes **Marks: 20**

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1. What is the formula for the half-way vector in the Blinn-Phong model? (1)

   A. $\frac{\vec{L}+\vec{V}}{2}$ B. $\frac{\vec{L}+\vec{H}}{\sqrt{2}}$ C. $2(\vec{L}\cdot\vec{N})\vec{N}-\vec{L}$ D. $\frac{\vec{L}+\vec{V}}{|\vec{L}+\vec{V}|}$ ✓

2. Ambient light is an approximation of — (1)

   A. Global illumination ✓ B. Path tracing C. Shadows D. Reflection

3. For the softest edges, we should use spot light falloff exponent $e=$? (1)

   A. 0 B. 2 C. 10 D. 20 ✓

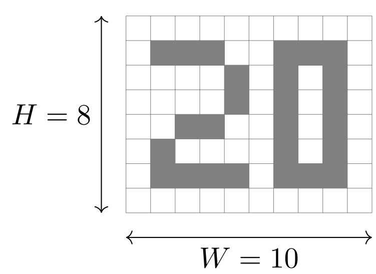
4. Which type of camera preserves distances? (1)

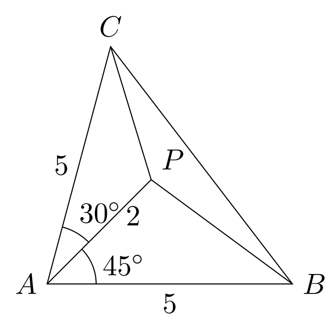
   A. Pinhole B. Simple field C. Fisheye D. Orthographic ✓

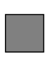
5. Which is not a feature of the simplified pinhole camera? (1)

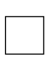
   A. Non-inverted image B. Perspective distortion ✓ C. Depth of field effect ✓ D. Unrealizable

6. Same triangle again! But now we want to texture it with an image. The (u,v) coordinates of the vertices A, B and C are $(0,0)$, $(1,0)$ and $(0.5,1)$ respectively.





 $(20,20,20)$

 $(200,200,200)$

a) What is the colour of P using nearest neighbour interpolation? (3)

Answer: From CT-3 we get :3,

$$\alpha = 0.5, \quad \beta = 0.207, \quad \gamma = 0.293$$

Therefore, the (u,v) coordinates of P are,

$$\begin{aligned} u &= \alpha\cdot 0+\beta\cdot 1+\gamma\cdot 0.5 = 0.3535 \\ v &= \alpha\cdot 0+\beta\cdot 0+\gamma\cdot 1 = 0.293 \end{aligned}$$

We also have, $w = W-1 = 9$ and $h = H-1 = 7$.

$$\begin{aligned} \text{color}(P) &= T\left[\text{round}(u\cdot w), \text{round}(v\cdot h)\right] \\ &= T\left[\text{round}(0.3535\cdot 9), \text{round}(0.293\cdot 7)\right] \\ &= T[3,2] = \text{rgb}(200,200,200) \end{aligned}$$

b) What is the colour of P using bilinear interpolation? (3)

We can see that the corner colours are:

$c_{00} = T\left[\lfloor u\cdot w\rfloor, \lfloor v\cdot h\rfloor\right] = T[3,2] = \text{rgb}(200,200,200)$

$c_{10} = T\left[\lfloor u\cdot w\rfloor+1, \lfloor v\cdot h\rfloor\right] = T[4,2] = \text{rgb}(200,200,200)$

$c_{01} = T\left[\lfloor u\cdot w\rfloor, \lfloor v\cdot h\rfloor+1\right] = T[3,3] = \text{rgb}(20,20,20)$

$c_{11} = T\left[\lfloor u\cdot w\rfloor+1, \lfloor v\cdot h\rfloor+1\right] = T[4,3] = \text{rgb}(200,200,200)$

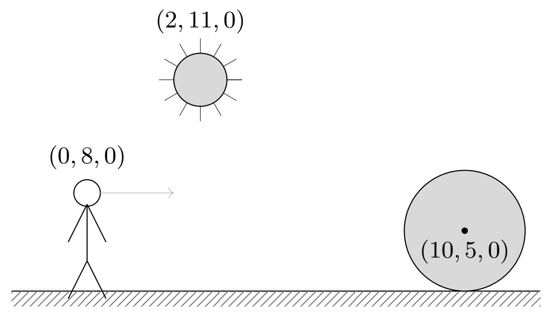
Now, we can calculate the interpolation factors:

$$\begin{aligned} t_u &= u\cdot w - \lfloor u\cdot w\rfloor = 0.3535\cdot 9 - 3 = 0.1805 \\ t_v &= v\cdot h - \lfloor v\cdot h\rfloor = 0.293\cdot 7 - 2 = 0.051 \end{aligned}$$

Therefore, the final colour is,

$$\begin{aligned} \text{color}(P) &= \text{lerp}(\text{lerp}(c_{00}, c_{10}, t_u), \text{lerp}(c_{01}, c_{11}, t_u), t_v) \\ &= \text{lerp}(\text{rgb}(200,200,200), \\ &\quad \text{lerp}(\text{rgb}(20,20,20), \text{rgb}(200,200,200), 0.1805), 0.051) \\ &= \text{lerp}(\text{rgb}(200,200,200), \text{rgb}(52.49,52.49,52.49), 0.051) \\ &\approx \text{rgb}(192.477,192.477,192.477) \\ &= \text{rgb}(192,192,192) \end{aligned}$$

7. As a game developer in Iron studio, you are working in a new level of a game. There is just a strange ball at $(10, 5, 0)$ of radius 5 and a point light at $(2, 11, 0)$.
After much thought, you have decided the following parameters using the Phong model:



- Point light intensity $\mathbf{I} = (20, 20, 20)$
- Ambient light $\mathbf{I}_a = (0.1, 0.1, 0.1)$
- Ambient Coefficient $\mathbf{k}_a = (0.1, 0.1, 0.1)$
- Diffuse Coefficient $\mathbf{k}_d = (0.4, 0.4, 0.4)$
- Specular Coefficient $\mathbf{k}_s = (0.5, 0.5, 0.5)$
- Shininess $n = 10$

But your game designer says the scene doesn't look right. She says the ball should be bluish gray in shadow [rgb(0,10,20)]. She also says the material should be less shiny. When the character at $(0, 8, 0)$ looks straight ahead (in the direction of x), he should see a gray colour [rgb(115,125,135)].

(**Hint:** Normalize RGB colours to $(0, 1)$ range.)

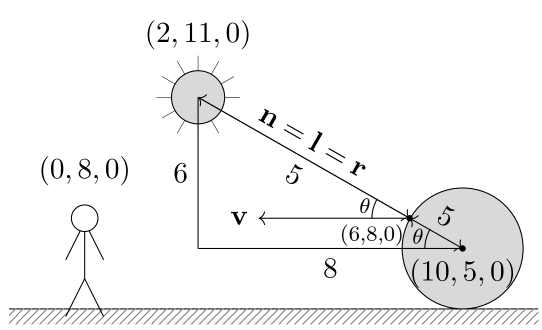
a) As per your game designer's requirement, what should be the value of the ambient coefficient $\mathbf{k}_a$? (3)

Answer: When the ball is in shadow, only ambient light contributes to the colour. Therefore,

$$\begin{aligned} \text{colour}_{\text{shadow}} &= \mathbf{I}_a \odot \mathbf{k}_a \\ \left(0, \frac{10}{255}, \frac{20}{255}\right) &= (0.1, 0.1, 0.1) \odot \mathbf{k}_a \\ \mathbf{k}_a = \left(0, \frac{10}{25.5}, \frac{20}{25.5}\right) &= (0, 0.392, 0.784) \end{aligned}$$

b) What should be the value of shininess n? (3)

Answer:



Now, to calculate the final colour, we need to know the angles or rather the dot products $\mathbf{n} \cdot \mathbf{l}$ and $\mathbf{r} \cdot \mathbf{v}$. Since the surface is a sphere, the normal points in the direction of the light. The light is also reflected in the same direction as the normal. So, $\mathbf{n} = \mathbf{l} = \mathbf{r}$ and $\mathbf{n} \cdot \mathbf{l} = 1$.

The angle between $\mathbf{v}$ and $\mathbf{n}$ is the angle θ which is the angle between the sides 10 and 8, in the right triangle with sides 6, 8 and 10 between the light and the sphere center. Therefore,

$$\mathbf{v} \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{r} = \cos\theta = \frac{8}{10} = 0.8$$

Also, we need to consider the light attenuation due to distance. The distance between the light and the point on the sphere is 5.

So, now th final colour is,

$$\begin{aligned} \text{colour} &= \mathbf{I}_a \odot \mathbf{k}_a + \frac{1}{d^2}\left[\mathbf{I} \odot \mathbf{k}_d(\mathbf{n} \cdot \mathbf{l}) + \mathbf{I} \odot \mathbf{k}_s(\mathbf{r} \cdot \mathbf{v})^n\right] \\ \left(\frac{115}{255}, \frac{125}{255}, \frac{135}{255}\right) &= \left(\frac{0}{255}, \frac{10}{255}, \frac{20}{255}\right) + \frac{1}{5^2}\left[(20, 20, 20) \odot (0.4, 0.4, 0.4)(1) + (20, 20, 20) \odot (0.5, 0.5, 0.5)(0.8)^n\right] \end{aligned}$$

Solving for the n in the R component, we get,

$$\begin{aligned} \frac{115}{255} &= 0 + \frac{1}{25}\left[8 + 10(0.8)^n\right] \\ (0.8)^n &= \frac{\frac{115 \cdot 25}{255} - 8}{10} = \frac{167}{510} \\ n &= \log_{0.8}\frac{167}{510} \approx 5.003 \approx 5 \end{aligned}$$

8. For your hard work and attending the exam so early in the morning. (3)
Answer: Thanks, I suppose.